

TESTs - Lesson 08 - Scripts - Ballot.sol

Why is it recommended to use a single private key for development purposes, rather than a full mnemonic key, if there's a risk of it being leaked?

Hardhat Runner

environment

configuration

development

Guides

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configuration

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A Private keys are easier to remember.

B A leaked private key only compromises one account, not the entire set of accounts linked to a mnemonic.

C Mnemonic keys are not compatible with Hardhat.

What is the primary conceptual distinction between using ``npx hardhat test`` and ``npx hardhat run`` commands for executing TypeScript files?

Test Suite

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A ``test`` compiles code, while ``run`` executes it without compilation.

B ``test`` is for executing test suites (e.g., Mocha blocks), while ``run`` is for executing scripts that perform operations (e.g., deploying contracts).

C ``test`` always targets a public blockchain, and ``run`` always targets a local one.

What does RPC stand for in the context of blockchain, and what is its main function when connecting to a public blockchain?

env

env.example

gitignore

mechardison

hardhat.config

package-lock.json

package.json

README.md

tsconfig.json

env

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tsconfig.json

A Remote Protocol Connection; for secure peer-to-peer communication.

B Random Program Code; for generating random transaction hashes.

C Remote Procedure Call; it allows interacting with a blockchain node without running one locally.

What is the primary limitation of Hardhat tasks (``npx hardhat run``) when trying to pass custom, dynamic arguments directly to a script from the command line?

environment

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A Hardhat only supports fixed, predefined arguments and does not recognize arbitrary positional arguments.

B Hardhat requires all arguments to be JSON objects.

C Hardhat is exclusively for local blockchain interactions and cannot process arguments for public testnets.

What are the essential steps to configure a Hardhat project for deploying a smart contract to a public testnet like Sepolia?



- ☒ A Add the testnet's network configuration (URL) to `hardhat.config.ts`, include an account (private key) in the network settings, and ensure API keys are secured in a `.env` file.
- ☐ B Install a new Hardhat plugin specifically for the Sepolia network and manually compile the smart contract for the testnet target.
- ☐ C Connect your MetaMask wallet directly to the Hardhat project and import the smart contract artifacts from Etherscan.

What is a significant advantage of directly using a library like `viem` for blockchain interaction over Hardhat's default tooling, especially for larger applications?



- ☒ A `viem` provides a lighter-weight and more direct way to connect to blockchains, allowing for greater control and reducing unnecessary dependencies compared to Hardhat's comprehensive environment.
- ☐ B `viem` offers built-in automatic code generation for contract ABIs and bytecode, which Hardhat lacks.
- ☐ C `viem` bypasses gas fees for all transactions, making development cheaper.

In the context of Hardhat and `viem`, what does 'ABI' stand for, and what role does it play when deploying and interacting with smart contracts?



- ☐ A Asynchronous Blockchain Interface; it ensures non-blocking contract calls.
- ☒ B Application Binary Interface; it defines how to interact with the compiled contract (function calls, arguments, return values).
- ☐ C Automated Bytecode Interpreter; it converts human-readable code into machine code.

Why is it crucial to keep your private keys and API keys secure, ideally in a `.env` file that is not committed to version control, and why should you avoid using mainnet keys for development?



- ☒ A To prevent unauthorized access to your funds and services, and to mitigate risks of financial loss or exploitation in case of a security breach.
- ☐ B Because `.env` files are automatically encrypted by Hardhat, making them safer than direct code inclusion.
- ☐ C Mainnet keys are not technically compatible with testnets, so they wouldn't work anyway.